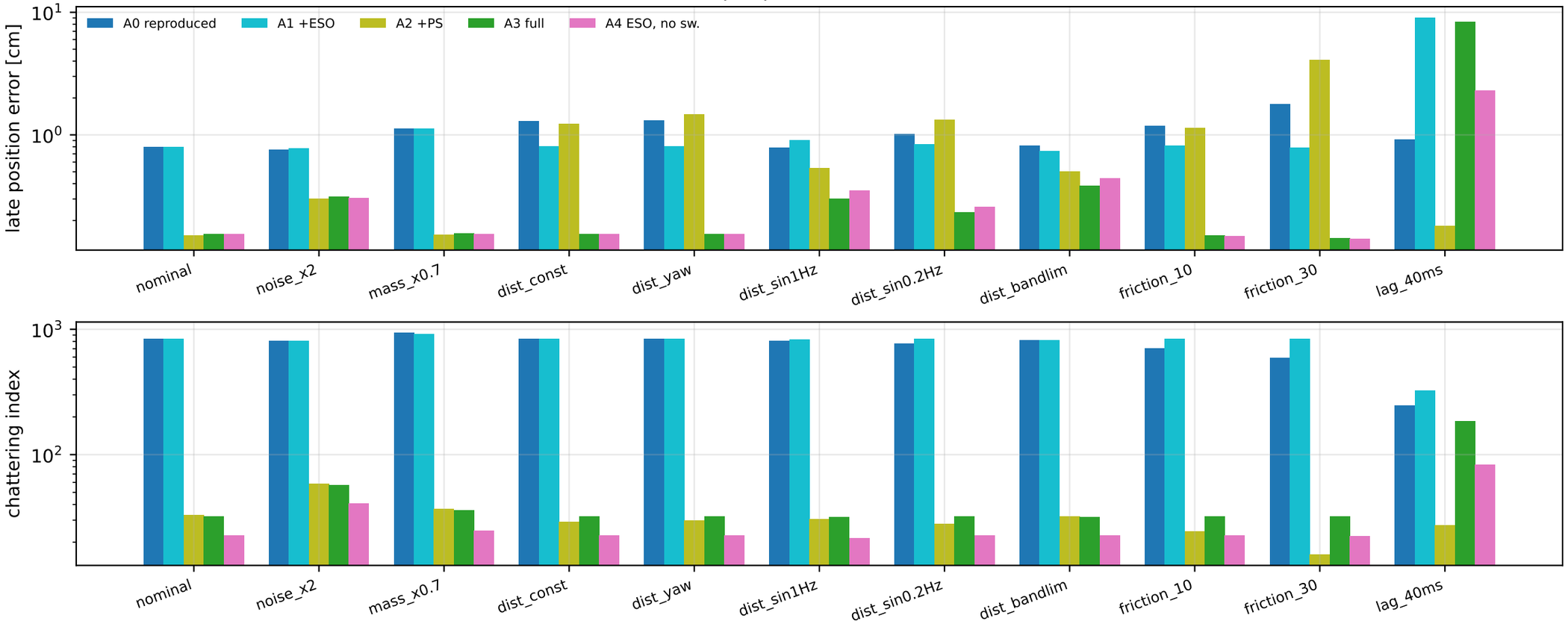
F6: pre-registered improvement — ablation A0–A4 and criteria

(a) ablation, torque plant, circle (mean of 5 seeds)



(c) pre-registered criteria P1–P7 (results/phase6/criteria.md)

	item	A3	threshold / reference	result
P1	circle dist_const late err [cm]	0.156 ± 0.021	≤ 0.5	PASS
P1	circle dist_yaw late err [cm]	0.156 ± 0.021	≤ 0.5	PASS
P1	circle friction_10 late err [cm]	0.151 ± 0.020	≤ 0.5	PASS
P1	circle friction_30 late err [cm]	0.143 ± 0.020	≤ 0.5	PASS
P1	figure8 dist_const late err [cm]	0.144 ± 0.025	≤ 0.5	PASS
P1	figure8 dist_yaw late err [cm]	0.144 ± 0.025	≤ 0.5	PASS
P1	figure8 friction_10 late err [cm]	0.140 ± 0.018	≤ 0.5	PASS
P1	figure8 friction_30 late err [cm]	0.152 ± 0.019	≤ 0.5	PASS
P2	circle dist_sin1Hz late err [cm]	0.303 ± 0.025	A0 0.782 ± 0.070 (+tol 0.140)	PASS
P2	circle dist_sin0.2Hz late err [cm]	0.233 ± 0.025	A0 1.012 ± 0.082 (+tol 0.163)	PASS
P2	circle dist_bandlim late err [cm]	0.384 ± 0.043	A0 0.819 ± 0.126 (+tol 0.252)	PASS
P2	figure8 dist_sin1Hz late err [cm]	0.299 ± 0.029	A0 0.691 ± 0.021 (+tol 0.059)	PASS
P2	figure8 dist_sin0.2Hz late err [cm]	0.222 ± 0.033	A0 0.863 ± 0.054 (+tol 0.108)	PASS
P2	figure8 dist_bandlim late err [cm]	0.377 ± 0.037	A0 0.665 ± 0.110 (+tol 0.220)	PASS
P3a	circle nominal chattering	32.2 ± 0.3	≤ 100	PASS
P3b	circle mass_x0.7 chattering	35.8 ± 0.4	≤ 2 × 32.2	PASS
P3b	circle noise_x2 chattering	56.8 ± 0.7	≤ 2 × 32.2	PASS
P3a	figure8 nominal chattering	31.5 ± 0.4	≤ 100	PASS
P3b	figure8 mass_x0.7 chattering	35.0 ± 0.5	≤ 2 × 31.5	PASS
P3b	figure8 noise_x2 chattering	55.4 ± 0.8	≤ 2 × 31.5	PASS
P4a	S1 ideal x_e settling [s]	0.35	≤ 1.36	PASS
P4b	circle nominal late err [cm]	0.156 ± 0.021	≤ A0 0.800	PASS
P4b	figure8 nominal late err [cm]	0.144 ± 0.025	≤ A0 0.762	PASS
P5	motor12V circle nominal late err [cm]	0.124 ± 0.023	≤ 2.0	PASS
P5	motor12V figure8 nominal late err [cm]	0.343 ± 0.031	≤ A0 6.056	PASS
P5 (report)	motor12V circle: A3 vs CKC+PI [cm]	0.124 ± 0.023	CKC+PI 1.331 ± 0.071	—
P5 (report)	motor12V figure8: A3 vs CKC+PI [cm]	0.343 ± 0.031	CKC+PI 13.732 ± 0.118	—
P6	ideal S3 shock-induced x_e [cm]	8.73	≤ A0 16.25	PASS
P6	motor12V S3 shock-induced x_e [cm]	5.68	≤ A0 6.09	PASS
P7	ĉ within [ĉ_min, ĉ_max] (grid A2/A3 + drift A3)	0 grid violations / 720 runs; 0 drift ticks	0	PASS
P7 (report)	time at ĉ_max, 200 s drift (A3, noisy, all plants/scenarios)	ĉ1 32%, ĉ2 35%, ĉ3 32%, ĉ4 63%	—	—